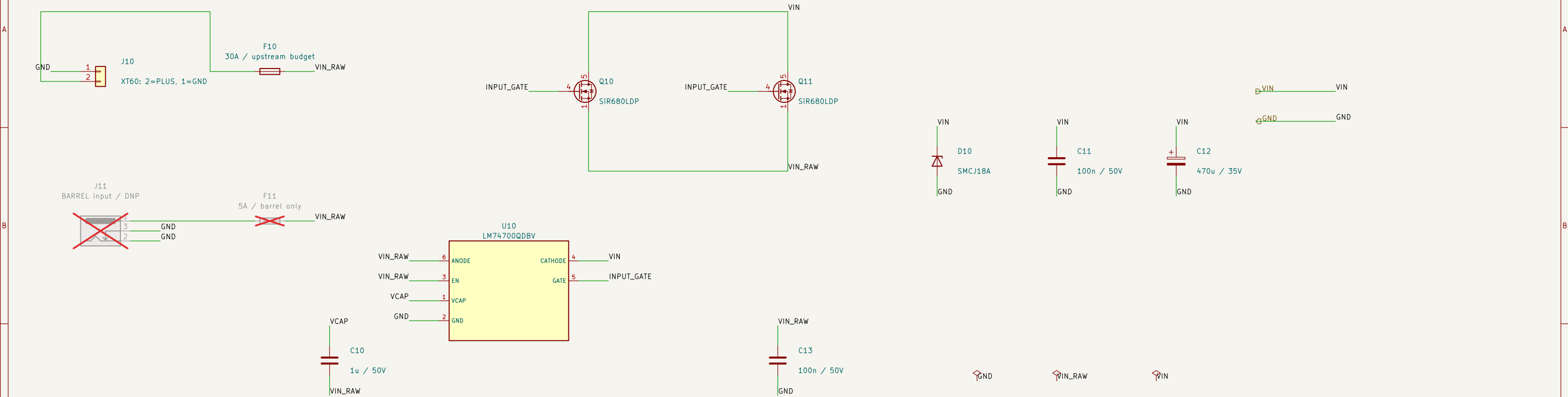


PULSARPOWER | 2 heaters + 4 monitored DC outlets + 4 USB 2.0 ports

PROTOTYPE FLOORPLAN – routing, full-load thermal validation and new firmware are still required.

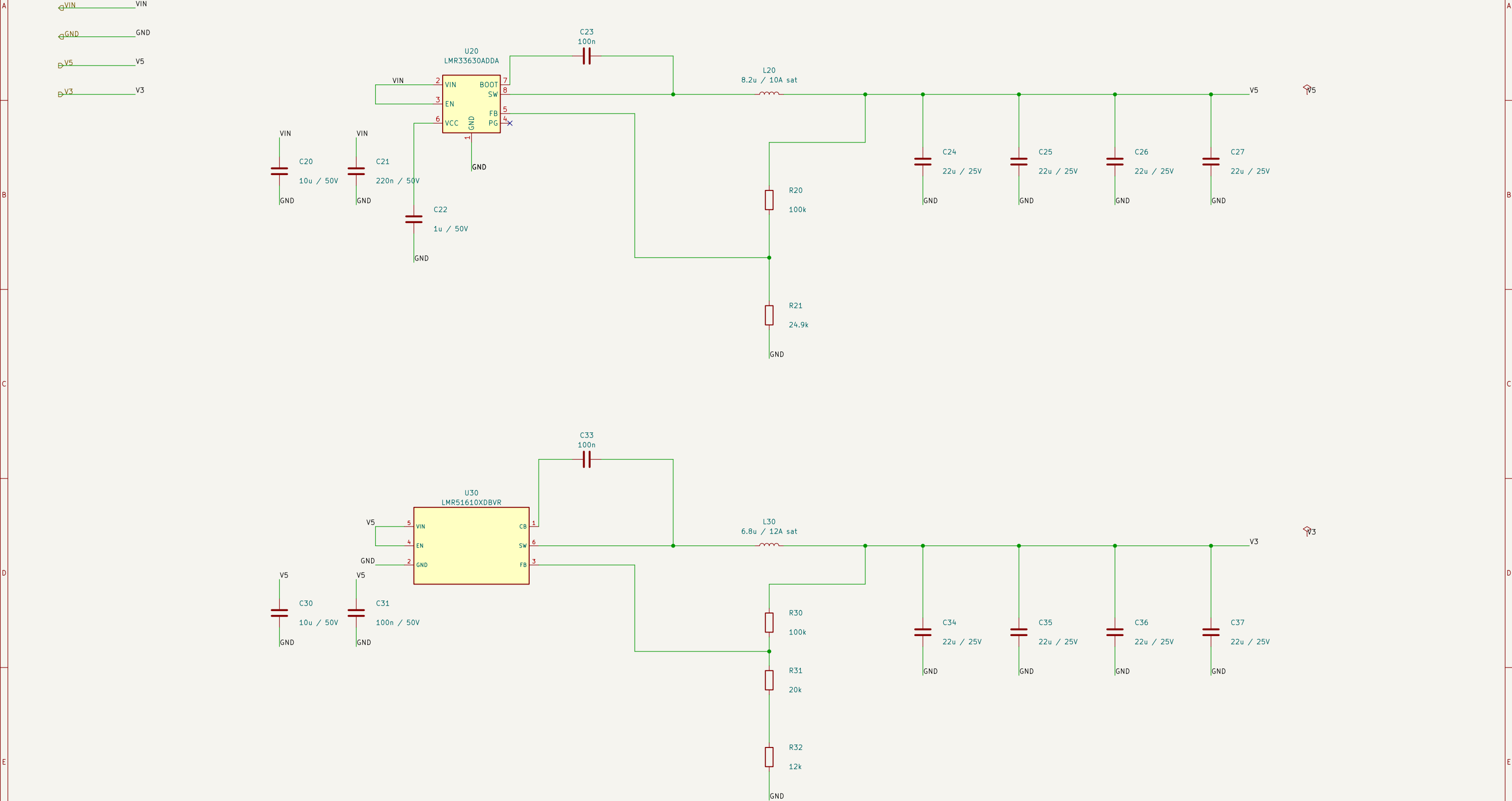


POWER INPUT | 12–18 V nominal | XT60 or barrel assembly option



FIT J10 + F10 OR J11 + F11. Do not populate both input connector paths.
XT60: provisional 25 A shared continuous budget; fuse, copper and temperature still require validation.
Barrel: 5 A absolute TOTAL ceiling; 3.5 A provisional continuous budget after fuse derating. Use a current-limited supply.
LM74700 + parallel Q10/Q11 provide reverse-polarity / reverse-current protection. It is not an overvoltage disconnect.
Never apply more than 18 V nominal. TVS pulse capability is not a sustained overvoltage rating.

DC SUPPLIES | 5 V / 3 A USB + 3.3 V / 650 mA logic budget



5 V: TI SNVSN3F Table 9–2; 8.2 uH nearest preferred value to 8 uH example.
3.3 V: TI SLUSEY1B; $0.8 \times (1 + 100k / (20k + 12k)) = 3.300$ V nominal. C32 removed (no VCC pin).
LMR51610X is 1 A / 400 kHz / PFM. This populated 6.8 uH stage has a 650 mA design load budget.
Place C31 at VIN/GND. C33 at CB/SW; keep the SW loop compact and R30/R31/R32 near FB.
USB budget: 4 x 500 mA plus 0.6 A for logic/support. Validate startup, MLCC bias, transients and temperature.

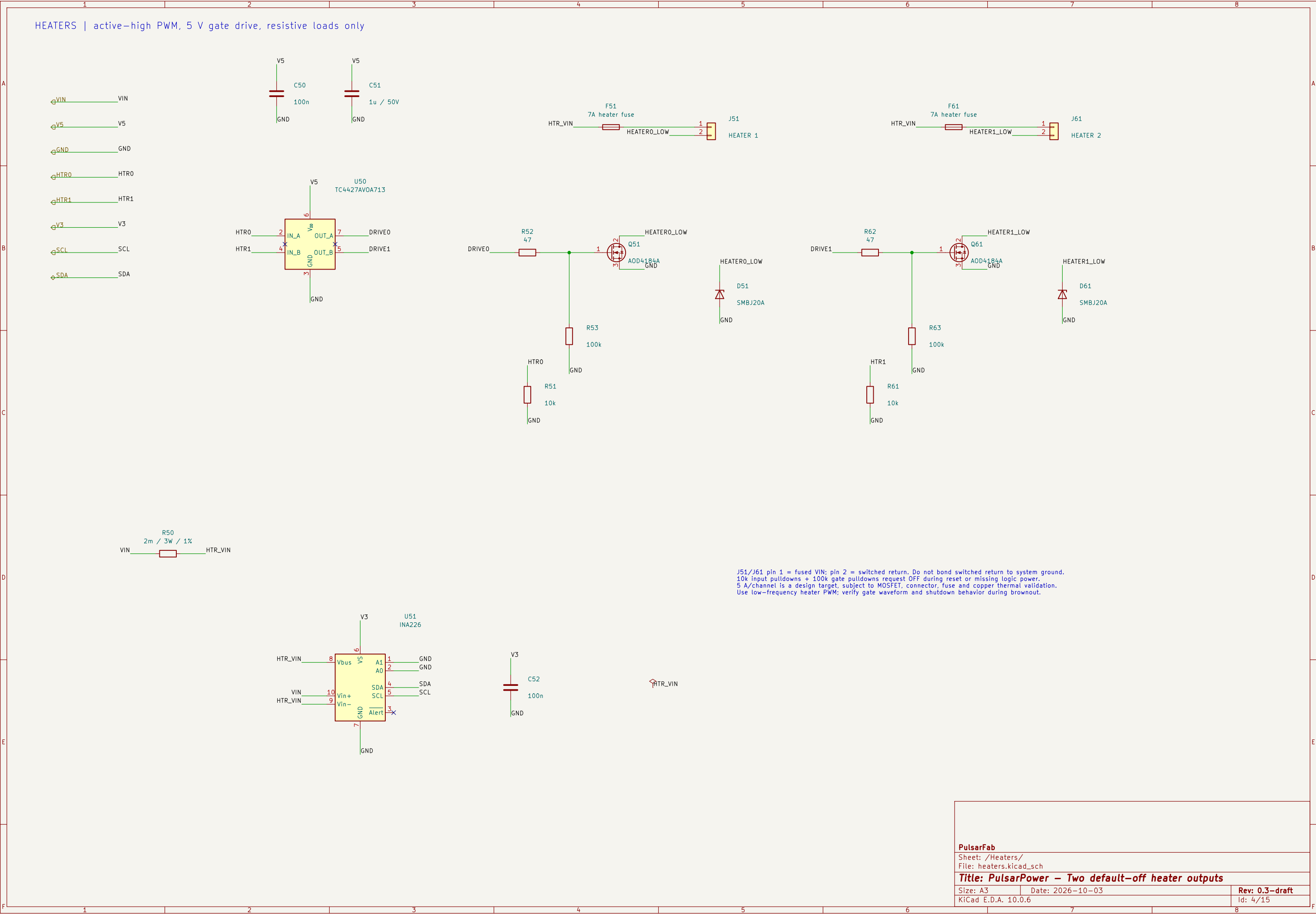
PulsarFab

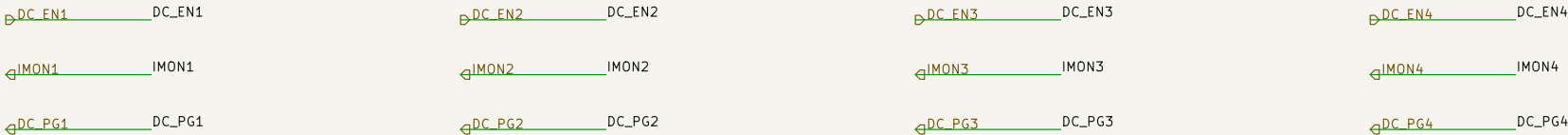
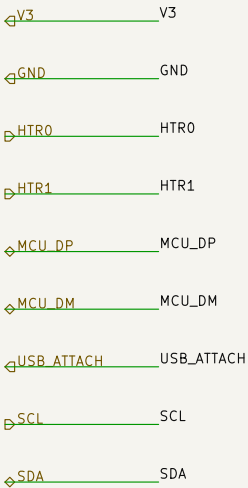
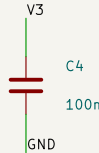
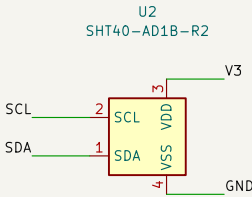
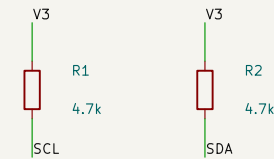
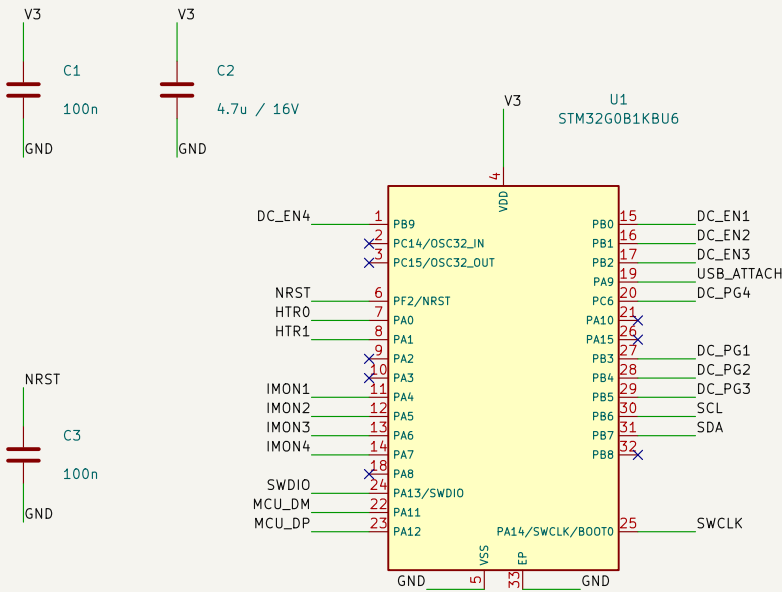
Sheet: /DC supplies/
File: supplies.kicad_sch

Title: PulsarPower – 5 V USB and 3.3 V logic supplies

Size: A3	Date: 2026–10–03	Rev: 0.3–draft
KiCad E.D.A. 10.0.6		Id: 3/15

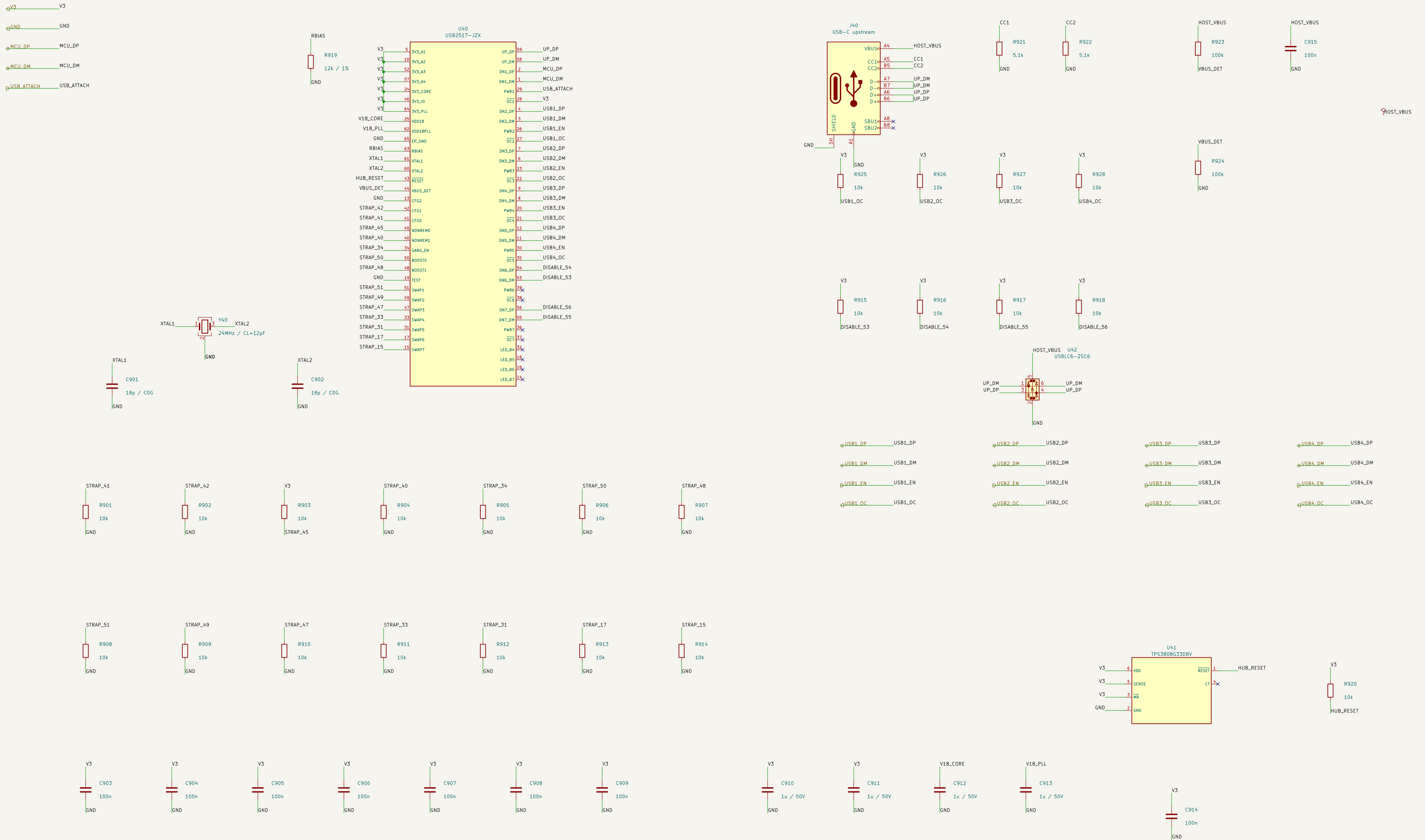
HEATERS | active-high PWM, 5 V gate drive, resistive loads only





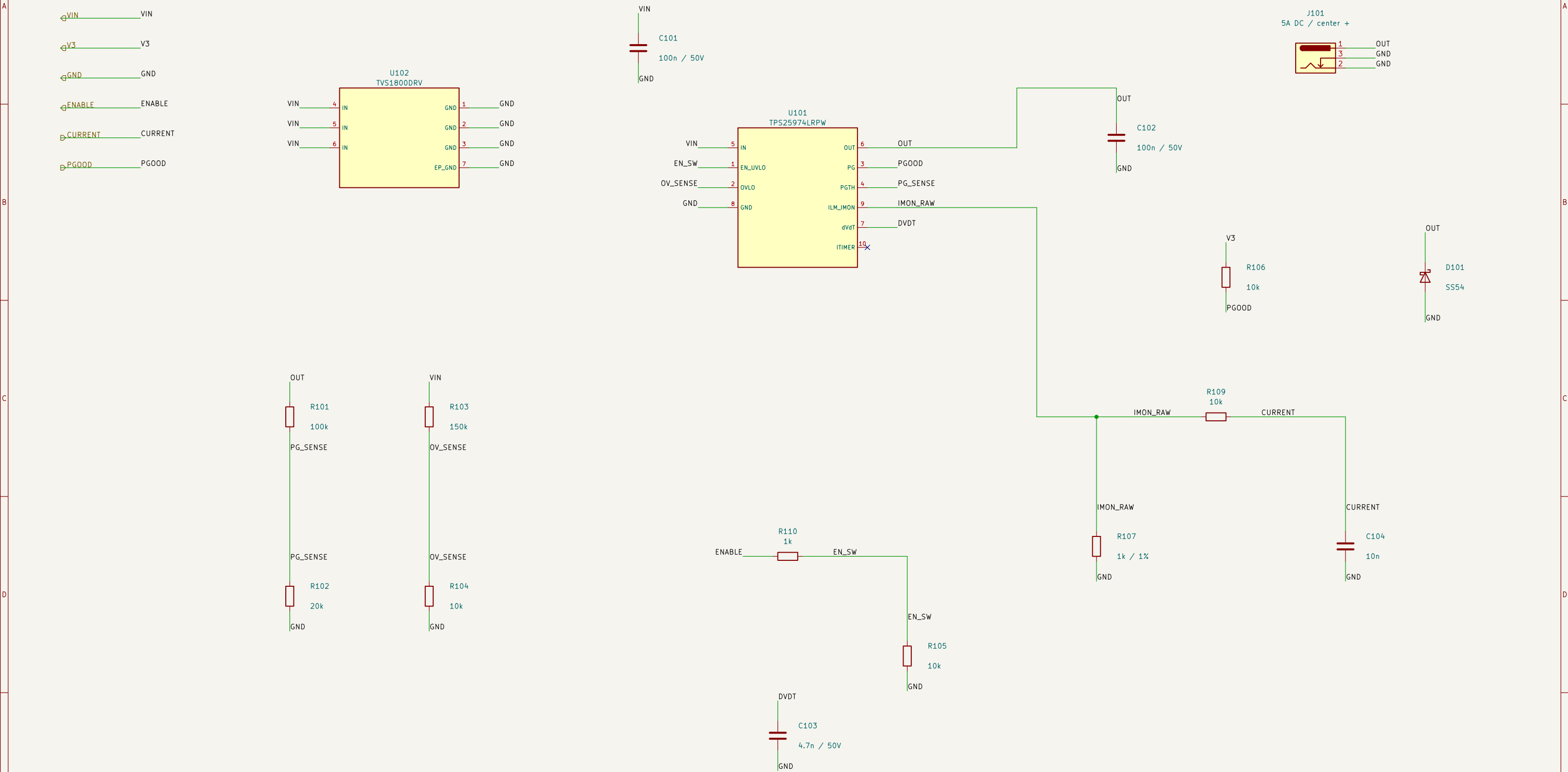
PA0/PA1: active-high TIM2 heater drive. PB0/PB1/PB2/PB9: static DC GPIO only (never PWM).
PA4..PA7: ADC IMON1..4. PB3/PB4/PB5/PC6: active-low DC faults.
PA9: internal hub port 1 enable (3.3 V). Attach USB only while high; disable UCPD dead-battery pull-down.
SHT40 at 0x44; keep physically away from power components. New firmware pin map is required.

USB HUB | USB2517 | internal port 1 + external ports 2-5; ports 6/7 disabled



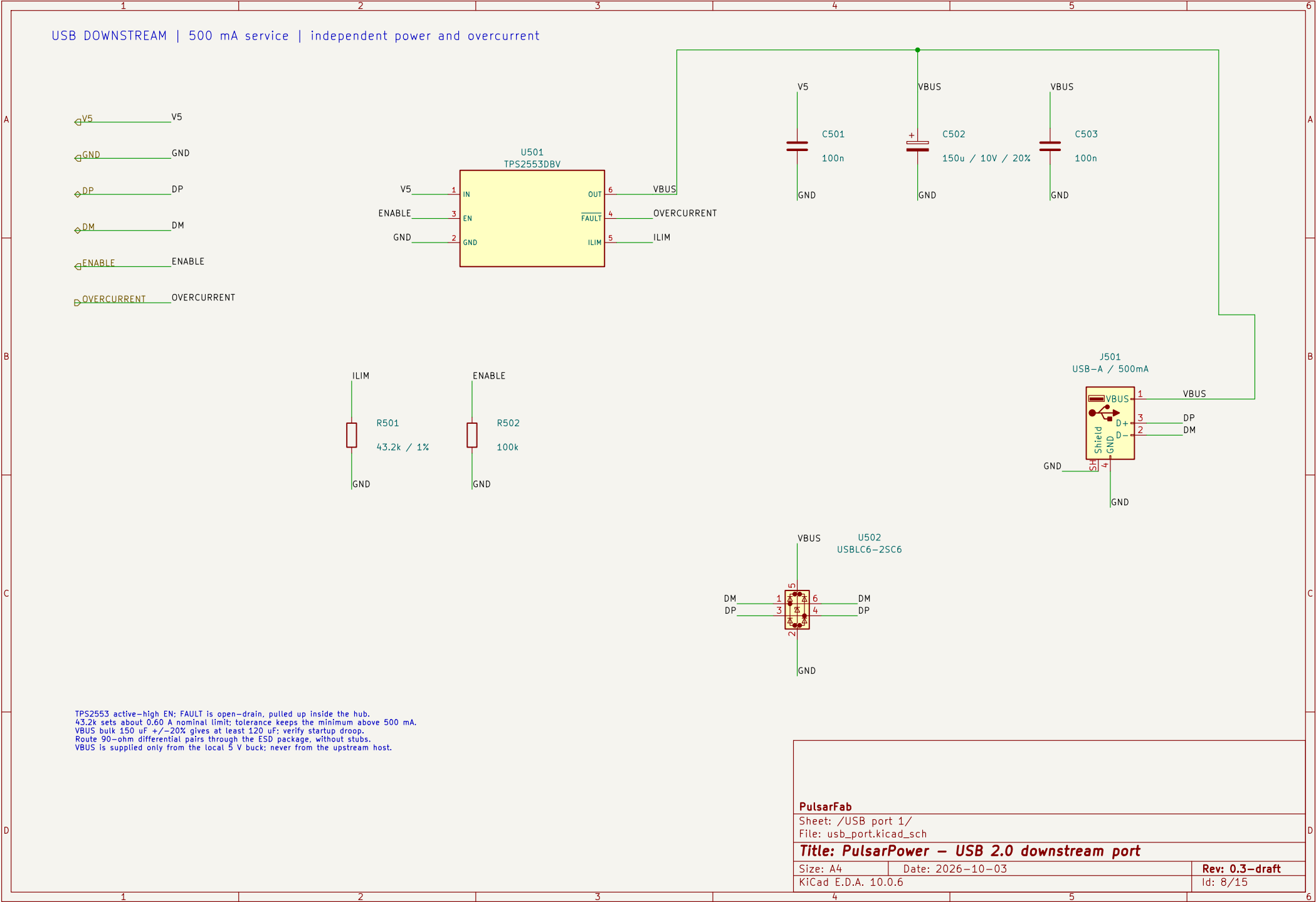
Straps: CFG=000 self-powered; NONREM=01 makes port 1 permanent; individual power/OC; normal polarity.
Unused ports 6/7; both D+ and D- pulled to 3V3 through 10k; PWR outputs unused, OC inputs unused.
VDD18 and VDD18PLL are separate INTERNAL regulator outputs; capacitors only; do not connect them together.
Supervisor: 3.07 V threshold, 20 ms delay. Crystal caps start at 18 pf; verify actual board parasitics/drive level.
Keep an uninterrupted ground plane under every USB pair. Verify 90-ohm geometry with the ordered stackup.

SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

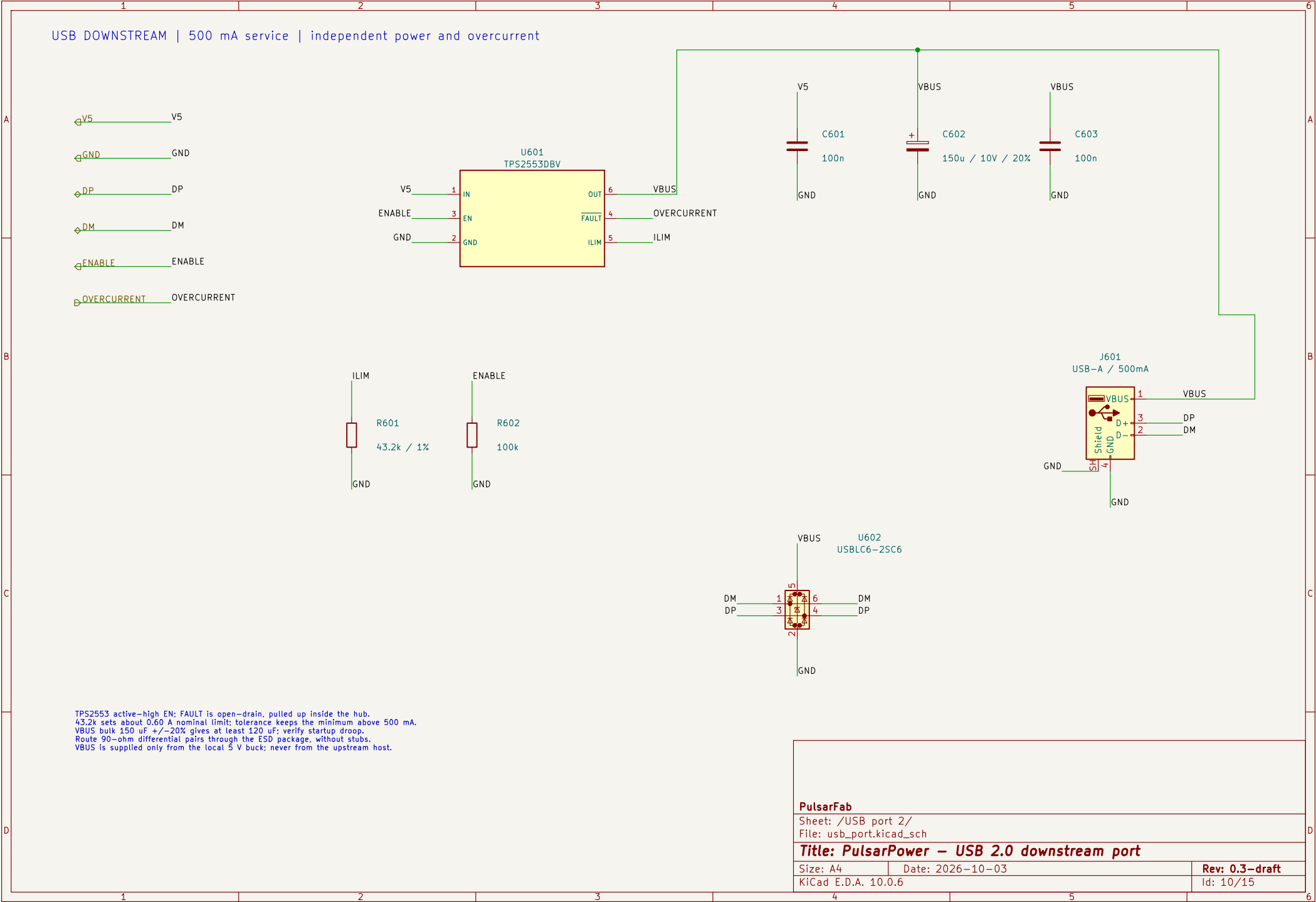


TPS25974L: 1k sets 5.75 A nominal breaker threshold for 5 A service. ITIMER open: latch-off.
ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
10k/10n filter isolates ADC capacitance from ILM. Calibrate channels; do not attach a capacitor directly to ILM.
Independent local OVLO: 150k/10k -> 19.2 V nominal. PG threshold: 100k/20k -> 7.2 V nominal.
Positive line switched; ground permanently connected. No reverse blocking; do not backfeed outputs.
IN pad5 / OUT pad6 are power/thermal rails, NOT GND. Reserve copper and thermal-via space.
TVS1800 flat clamp replaces SMBJ18A: 24.7 V max at specified 35 A / 8–20 us / 125 C pulse; verify loop overshoot.

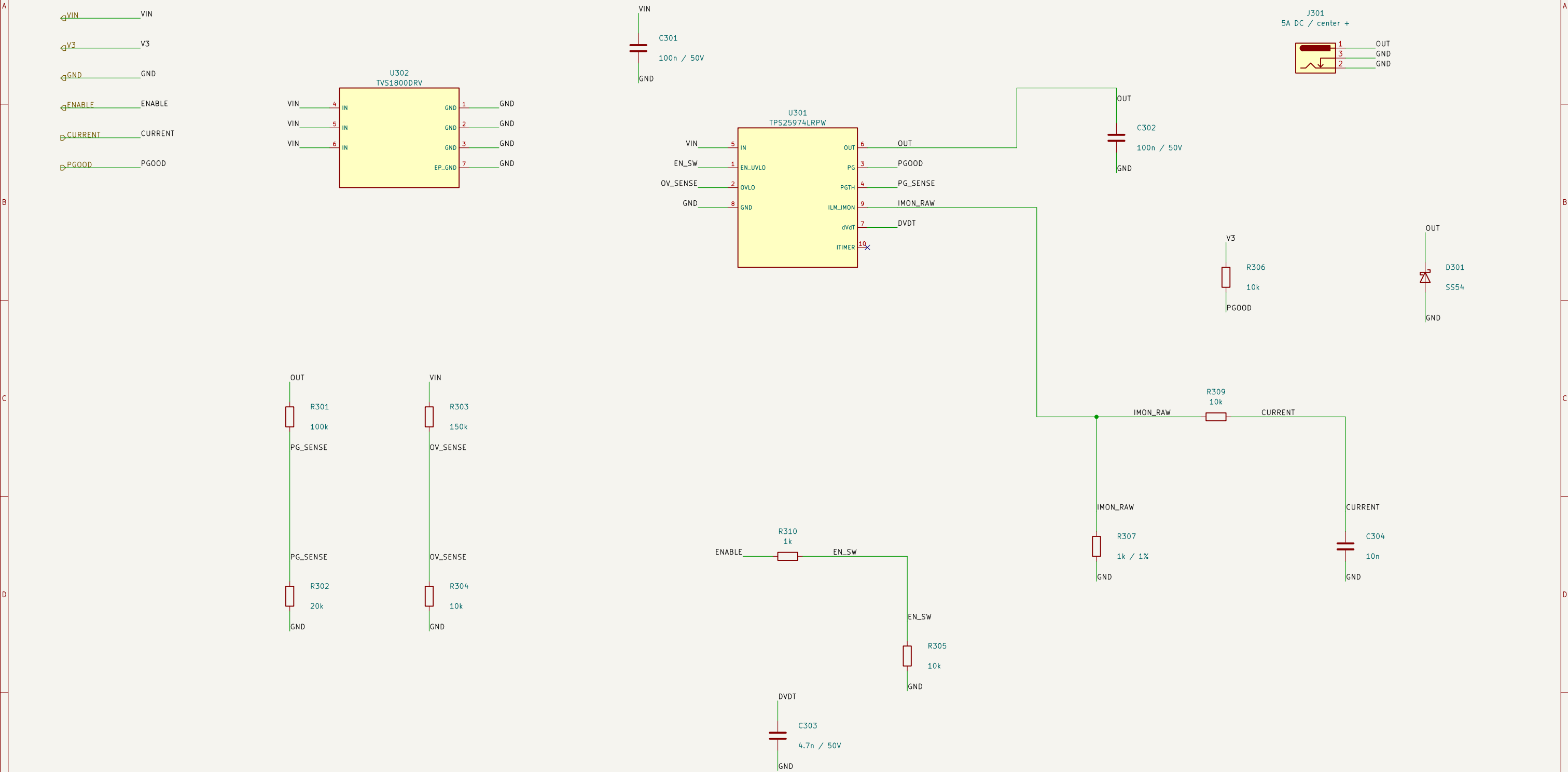
USB DOWNSTREAM | 500 mA service | independent power and overcurrent



USB DOWNSTREAM | 500 mA service | independent power and overcurrent

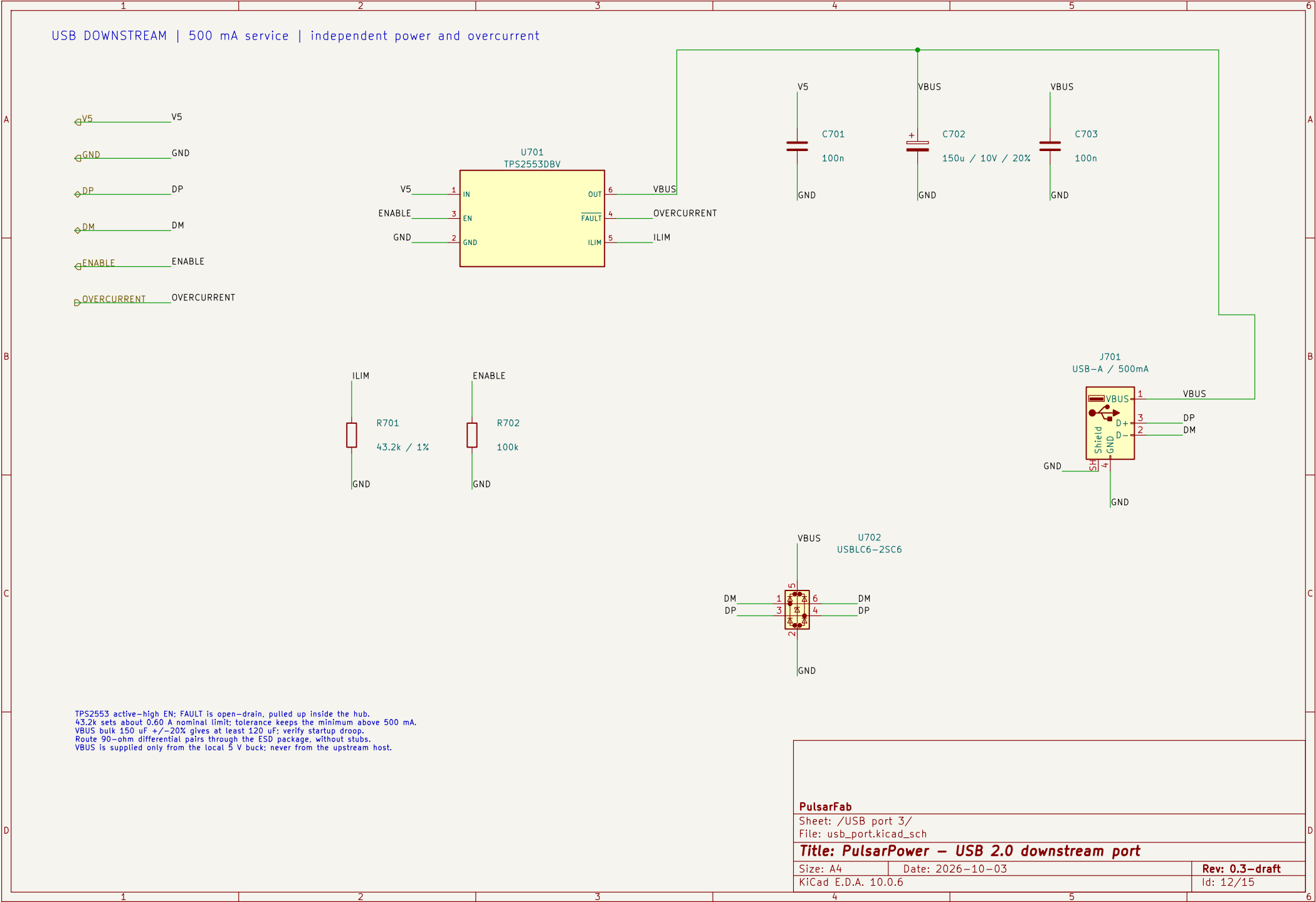


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

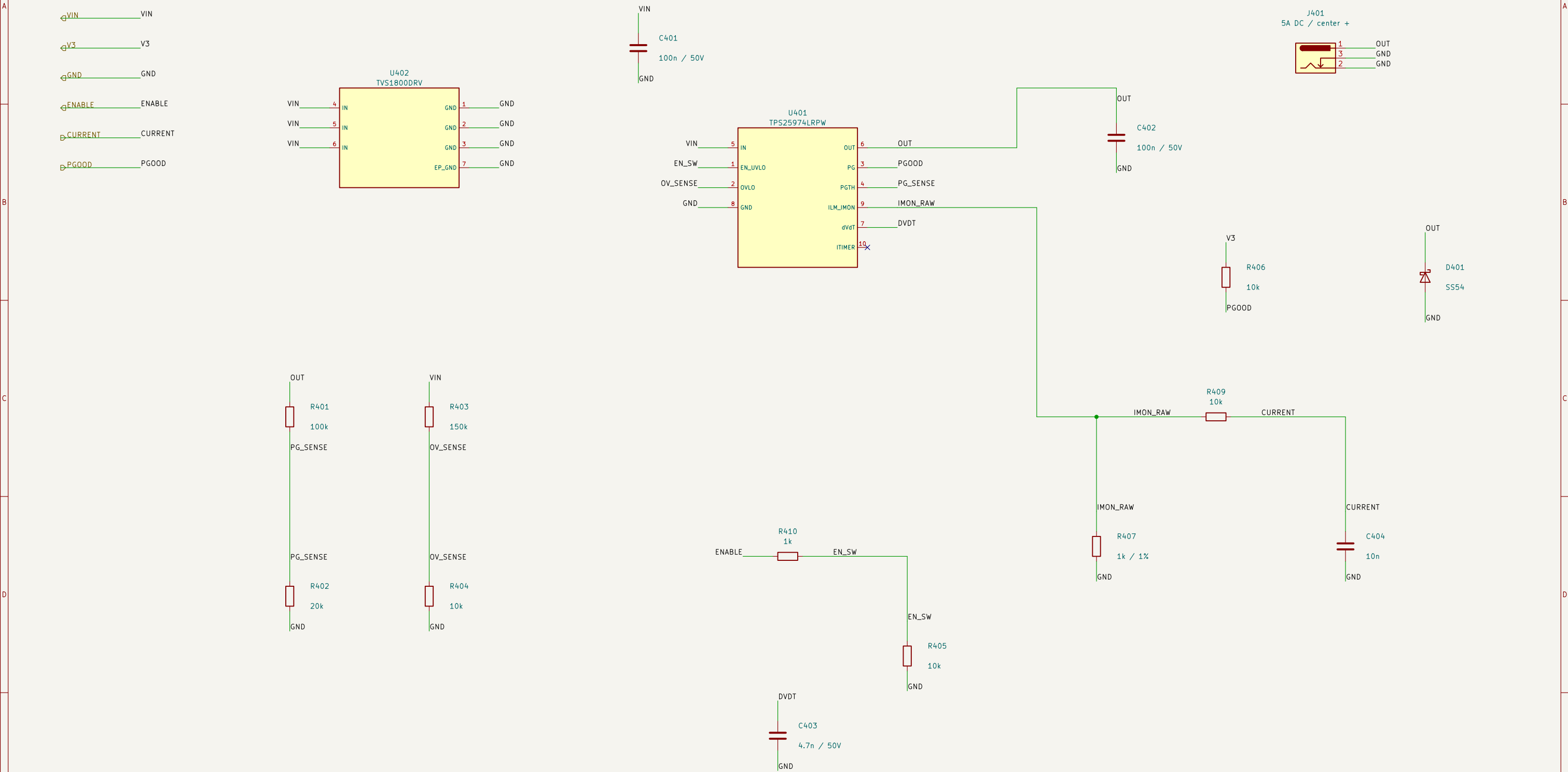


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ILM is BOTH current setting and monitoring: 105.5 uA/A x 1k = 0.1055 V/A; 0.528 V at 5 A.
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USB DOWNSTREAM | 500 mA service | independent power and overcurrent

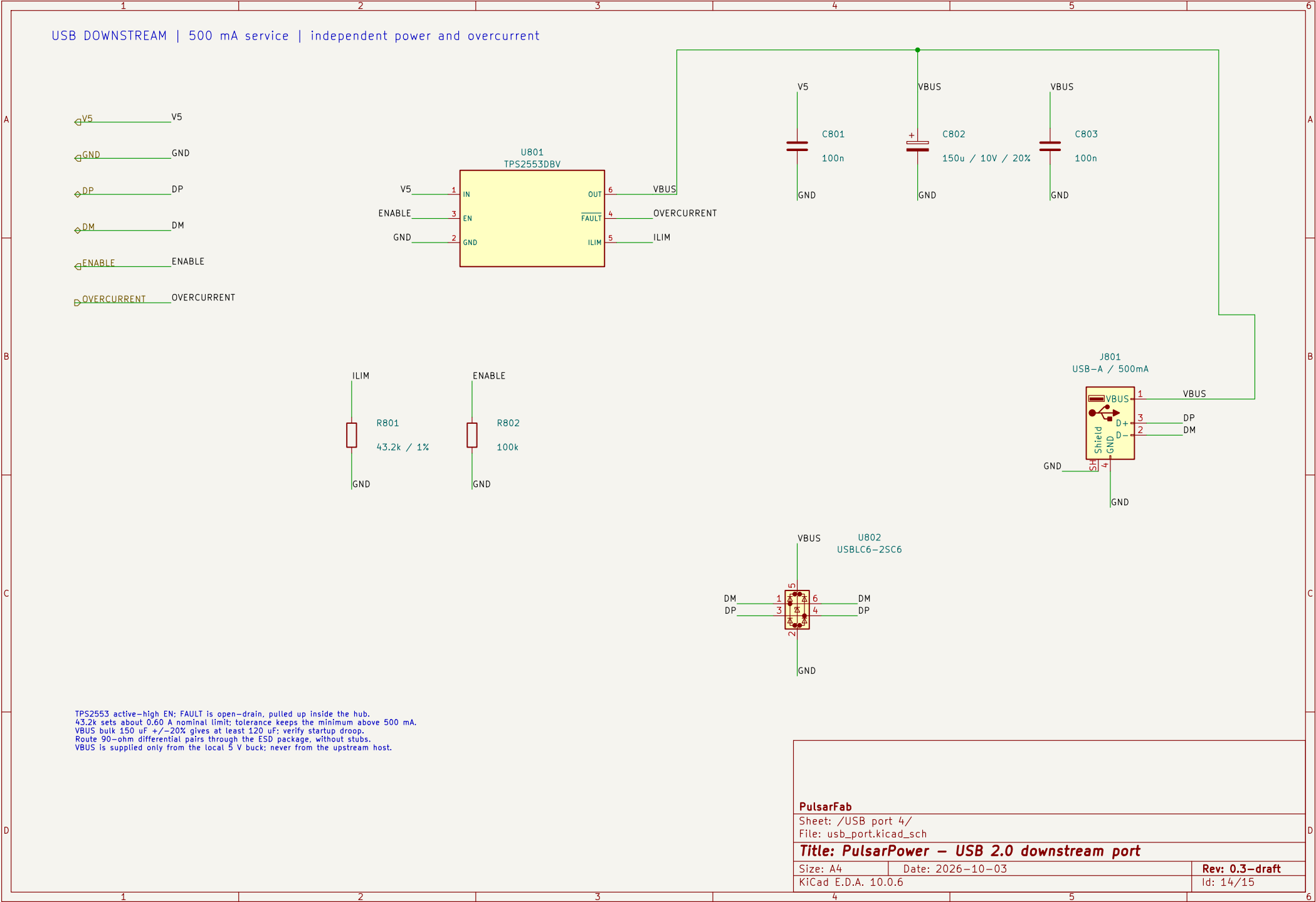


SWITCHED DC OUTLET | 12–18 V / 5 A service | hardware voltage inhibit / latched breaker

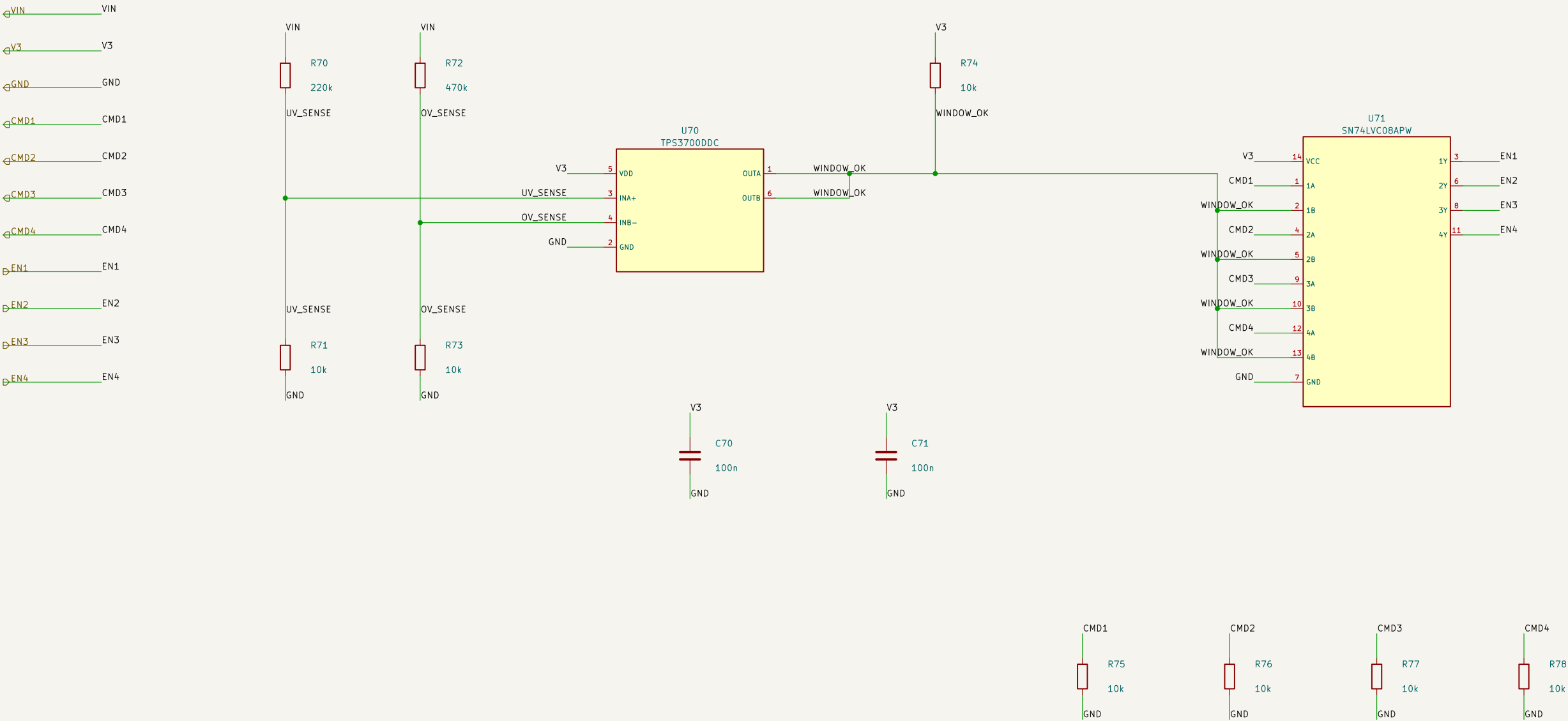


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USB DOWNSTREAM | 500 mA service | independent power and overcurrent



DC PROTECTION | one hardware voltage window | four independent default-off commands



TPS3700: 9.2 V UV rising / 19.2 V OV rising nominal. Open-drain outputs wired together.
74LVC08: ENi = CMDi AND WINDOW_OK; each command has a 10k pulldown.
Each outlet also has an EN pulldown: reset, missing logic supply and disconnected command request OFF.
A voltage recovery can re-enable an asserted command and reset a latched breaker; firmware must deassert on unexpected PG loss.
Shared inhibit is a common failure point, not a redundant safety channel. Local eFuse OVLO remains independent.
This circuit does not disconnect VIN from the regulator, heater stages or eFuse input pins.